

Chapter 15- EIGRP Unequal Cost, Loopback Interface

1 Unequal Cost Load Balancing in EIGRP

What is Unequal Cost Load Balancing?

Most routing protocols (like RIP, OSPF) support **equal-cost load balancing only**.

But **EIGRP is special** 

It supports **Unequal Cost Load Balancing** using the **variance command**.

 That means traffic can be shared between routes that have **different metrics**.

Why is it Useful?

Imagine:

- One road is 4 lanes (fast)
- Another road is 2 lanes (slower)

Normally, routers use only the best road.

But EIGRP can use **both roads** intelligently based on their capacity.

This improves:

-  Bandwidth usage
 -  Network performance
 -  Redundancy
-



Terms in EIGRP

EIGRP uses **DUAL (Diffusing Update Algorithm)**.

◆ 1. Feasible Distance (FD)



Definition:

Total metric from **your router to destination**.

It includes:

- Cost from your router to neighbor
- Neighbor's cost to destination

Example:

- Cost to neighbor = 20
- Neighbor says cost to destination = 30

$FD = 20 + 30 = 50$



FD is installed in routing table.

◆ 2. Reported Distance (RD)

(Also called Advertised Distance)



Definition:

The metric your neighbor reports to reach destination.

In above example:

$RD = 30$

$FD = 50$

◆ 3. Successor

🏆 Best path (lowest FD).

Installed in routing table.

Every reachable network must have a successor.

◆ 4. Feasible Successor (Backup Route)

📁 Backup path stored in topology table.

Used immediately if successor fails.

🔒 Feasibility Condition (Very Important)

A route becomes feasible successor only if:

$RD < FD$ of current successor

This prevents routing loops 🔄

📁 How Variance Works

Default:

variance 1

Means → Only equal-cost paths used.

📌 Formula

A path is eligible if:

$\text{Route Metric} \leq (\text{Best FD} \times \text{Variance})$

Example:

Path FD

A 100

B 120

C 200

If Variance = 1

Eligible:

≤ 100

Only Path A

If Variance = 2

$100 \times 2 = 200$

Eligible:

A, B, C

Now EIGRP can use all 3 paths 🎉

🔧 How to Configure Variance

On router:

```
router eigrp 100
```

```
variance 2
```

That's it ✅

Traffic Sharing

By default:

traffic-share balanced

Traffic is shared proportionally based on metric.

Loopback Interface (Virtual Interface)

What is Loopback Interface?

Loopback is:

- Virtual interface
 - Always UP (unless shutdown)
 - Not connected to physical cable
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Why We Create Loopback?

Used for:

- Router ID
 - Testing
 - Stability
 - Management
 - Simulation in labs
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Real-Life Example

Think of loopback like:

 Your permanent home address

Even if roads close, your home address stays same.



Router ID Selection Rules (3 Rules)

- 1 Highest IP on Loopback interface
- 2 If no loopback → Highest IP on physical interface
- 3 If manually configured → That value used

Loopback is preferred because:

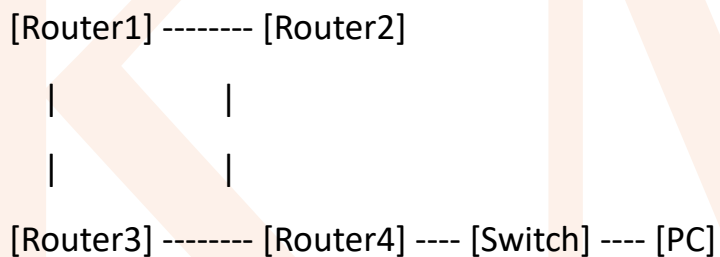
- Always stable
- Never goes down



EIGRP LOOPBACK LAB (4 Routers)



Topology



IP Addressing Plan

Link	Network
------	---------

R1–R2	10.0.12.0/30
-------	--------------

R1–R3	10.0.13.0/30
-------	--------------

R2–R4	10.0.24.0/30
-------	--------------

R3–R4	10.0.34.0/30
-------	--------------

LAN:

192.168.4.0/24 (PC Network)

Q1.Router Configurations (Step-by-Step)

◆ Router 1

hostname R1

int g0/0

ip address 10.0.12.1 255.255.255.252

no shutdown

int fa1/0

ip address 10.0.13.1 255.255.255.252

no shutdown

◆ Router 2

hostname R2

int g0/0

ip address 10.0.12.2 255.255.255.252

no shutdown

int fa1/0

ip address 10.0.24.1 255.255.255.252

no shutdown

◆ Router 3

hostname R3

int fa1/0

ip address 10.0.13.2 255.255.255.252

no shutdown

int fa2/0

ip address 10.0.34.1 255.255.255.252

no shutdown

◆ Router 4

hostname R4

int fa2/0

ip address 10.0.34.2 255.255.255.252

no shutdown

int fa1/0

ip address 10.0.24.2 255.255.255.252

no shutdown

int g0/0

ip address 192.168.4.254 255.255.255.0

no shutdown



PC Configuration

IP: 192.168.4.1

Mask: 255.255.255.0

Gateway: 192.168.4.254

Q2.Configure Loopback Interfaces

R1

int loopback0

ip address 1.1.1.1 255.255.255.255

R2

int loopback0

ip address 2.2.2.2 255.255.255.255

R3

int loopback0

ip address 3.3.3.3 255.255.255.255

R4

int loopback0

ip address 4.4.4.4 255.255.255.255

Q3.Configure EIGRP (Disable Auto Summary)

◆ Router 1

```
router eigrp 100
no auto-summary
network 10.0.12.0 0.0.0.3
network 10.0.13.0 0.0.0.3
network 1.1.1.1 0.0.0.0
passive-interface loopback0
```

◆ Router 2

```
router eigrp 100
no auto-summary
network 10.0.12.0 0.0.0.3
network 10.0.24.0 0.0.0.3
network 2.2.2.2 0.0.0.0
passive-interface loopback0
```

◆ Router 3

```
router eigrp 100
no auto-summary
network 10.0.13.0 0.0.0.3
network 10.0.34.0 0.0.0.3
network 3.3.3.3 0.0.0.0
passive-interface loopback0
```

◆ **Router 4**

```
router eigrp 100
no auto-summary
network 10.0.24.0 0.0.0.3
network 10.0.34.0 0.0.0.3
network 192.168.4.0 0.0.0.255
network 4.4.4.4 0.0.0.0
passive-interface g0/0
passive-interface loopback0
```

Q4. Configure Unequal Cost Load Balancing on R1

We want R1 to load balance toward 192.168.4.0/24.

R1 has 2 paths:

- Via R2
 - Via R3
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Step 1: Check Metrics

show ip eigrp topology

Find FD of best path.

Step 2: Configure Variance

If best FD = example 10000

Second path = 15000

Then:

```
router eigrp 100
```

```
variance 2
```

Now R1 will use both paths.



Commands to Verify Lab Success



Check Neighbors

show ip eigrp neighbors



Check Routing Table

show ip route

Look for:

D 192.168.4.0



Check Topology Table

show ip eigrp topology



Check Load Balancing

show ip route 192.168.4.0

You should see multiple next-hops.



Ping Test

From R1:

ping 192.168.4.1

From PC:

ping 1.1.1.1

 Final Summary

EIGRP Advanced Concepts:

- ✓ Supports unequal-cost load balancing
 - ✓ Uses FD and RD
 - ✓ Uses DUAL
 - ✓ Loopback improves stability
 - ✓ Variance controls traffic sharing
 - ✓ Passive interface improves security
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1 DETAILED SUMMARY



1 Unequal Cost Load Balancing in EIGRP



What is Unequal Cost Load Balancing?

Most routing protocols like:

- RIP
- OSPF

Support only **Equal-Cost Load Balancing**.

But **EIGRP** is special 💎

It supports **Unequal Cost Load Balancing** using:

variance command



It allows traffic sharing between routes with different metrics.



Why It Is Useful?

Imagine:

- 🚧 Road A = 4 lanes (fast)
- 🚧 Road B = 2 lanes (slower)

Normally → Only best path used.

With EIGRP:

- ✓ Both paths can be used
 - ✓ Traffic shared intelligently
 - ✓ Better bandwidth utilization
 - ✓ Higher redundancy
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Important EIGRP Terms

EIGRP uses **DUAL (Diffusing Update Algorithm)** 

◆ **1 Feasible Distance (FD)**

 Definition:

Total metric from your router to destination.

It includes:

- Cost to neighbor
- Neighbor's cost to destination

Example:

- Cost to neighbor = 20
- Neighbor's cost = 30

$$FD = 20 + 30 = 50$$

 Installed in routing table.

◆ **2 Reported Distance (RD)**

Also called:

- Advertised Distance

 Definition:

Distance reported by neighbor to destination.

In above example:

- RD = 30
 - FD = 50
-

◆ **3 Successor**

- Best path (lowest FD)

- Installed in routing table
 - Every reachable network must have one
-

◆ Feasible Successor

- Backup path
 - Stored in topology table
 - Used immediately if successor fails
-

Feasibility Condition (Very Important)

A route becomes Feasible Successor if:

$RD < FD$ (of current successor)

- ✓ Prevents routing loops
 - ✓ Ensures loop-free backup
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How Variance Works

Default:

variance 1

Means:

Only equal-cost paths allowed.

Eligibility Formula

A path is used if:

$\text{Route Metric} \leq (\text{Best FD} \times \text{Variance})$

Example:

Path FD

A 100

B 120

C 200

If variance = 1

Eligible:

≤ 100

Only Path A

If variance = 2

$100 \times 2 = 200$

Eligible:

A, B, C 🎉

Now EIGRP can load balance across all.

⚙️ Configure Variance

router eigrp 100

variance 2

Done ✅

🔄 Traffic Sharing

Default:

traffic-share balanced

Traffic shared proportionally based on metric.

Loopback Interface

What is Loopback?

Loopback is:

- Virtual interface
- No physical cable
- Always UP (unless shutdown)
- Logical interface

Why Use Loopback?

Used for:

- Router ID
- Stability
- Testing
- Management
- Lab simulation

Real-Life Example

Loopback = Permanent home address 

Even if roads close → Address remains same.

Router ID Selection Rules

Priority Order:

- 1 Highest IP on Loopback
- 2 Highest IP on Physical Interface
- 3 Manually configured Router ID

Loopback preferred because:

- ✓ Stable
- ✓ Never goes down

LAB OVERVIEW

Topology:

R1 — R2

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R3 — R4 — PC

Networks:

- 10.0.12.0/30
- 10.0.13.0/30
- 10.0.24.0/30
- 10.0.34.0/30
- 192.168.4.0/24

Loopbacks:

- 1.1.1.1
 - 2.2.2.2
 - 3.3.3.3
 - 4.4.4.4
-



COMPLETE MIND MAP

EIGRP Advanced

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|— Unequal Cost Load Balancing

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|— Variance

|

|— Eligibility formula

|

|— Traffic-share balanced

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|— EIGRP Terms

|

|— FD

|

|— RD

|

|— Successor

|

|— Feasible Successor

|

|— Feasibility Condition

|

|— Variance Example

|

|— Loopback Interface

|

|— Virtual

|

|— Always UP

|

|— Router ID

|

|— Stability

|

|— Router ID Rules

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- └─ Lab
 - └─ 4 Routers
 - └─ Loopbacks
 - └─ Disable auto-summary
 - └─ Verification
-

Q & A

Q 1 What is Unequal Cost Load Balancing?

Answer:

It allows EIGRP to use multiple paths with different metrics using variance command.

Q 2 Which routing protocol supports unequal cost load balancing?

Answer:

EIGRP supports unequal-cost load balancing.

Q 3 What is variance in EIGRP?

Answer:

Variance is a multiplier that allows routes within certain metric range to be used for load balancing.

Q 4 What is default variance?

Answer:

Default value is 1 (equal-cost only).

Q 5 Explain Feasible Distance (FD).

Answer:

FD is total metric from current router to destination.

Q 6 Explain Reported Distance (RD).

Answer:

RD is distance reported by neighbor to reach destination.

Q 7 What is Successor?

Answer:

Best path selected by DUAL, installed in routing table.

Q 8 What is Feasible Successor?

Answer:

Backup path that satisfies feasibility condition.

Q 9 What is Feasibility Condition?

Answer:

$RD < FD$ of current successor.

It prevents routing loops.

Q 10 What happens if RD is greater than FD?

Answer:

It cannot become feasible successor.

Q 1 1 How does EIGRP prevent loops?

Answer:

Using DUAL algorithm and feasibility condition.

Q 1 2 What is Loopback Interface?

Answer:

Virtual interface that is always up and used for router ID and stability.

Q 1 3 Why is loopback preferred for Router ID?

Answer:

Because it never goes down, so router ID remains stable.

Q 1 4 Router ID selection order?

Answer:

1. Highest Loopback IP
 2. Highest Physical IP
 3. Manually configured
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Q 1 5 What is traffic-share balanced?

Answer:

Traffic is shared proportionally according to metric.

Q 1 6 Why disable auto-summary?

Answer:

To prevent classful summarization and allow proper routing in discontinuous networks.

Q 1 7 How to verify load balancing?

Answer:

Use:

show ip route

show ip eigrp topology

Multiple next-hops should appear.

🔲 FINAL CONCLUSION

- 🔥 EIGRP is powerful because it supports unequal cost load balancing.
 - 🔥 Variance allows intelligent traffic sharing.
 - 🔥 FD and RD are core concepts for interviews.
 - 🔥 Feasibility condition prevents loops.
 - 🔥 Loopback interface ensures stability and proper Router ID selection.
 - 🔥 Advanced EIGRP makes enterprise networks efficient and redundant.
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